

CIS 476 Term Project

Document/Program and Demo Due Dates

Monday 04/13/26

This project is required for all students. You may finish the project all by yourself or team with others. 3 team members maximum.

In this project, you are required to design DriveShare - Peer-to-Peer Car Rental Platform. DriveShare is a standalone web or mobile platform designed to connect car owners with individuals seeking short-term car rentals. The platform aims to provide a convenient and efficient solution for users to rent personal vehicles directly from their owners. DriveShare facilitates listing, searching, transactions and enhances user communication to ensure a seamless car-sharing experience. Use Turo.com as an example.

Functions/Features

- User Registration and Authentication:
 - Users can register on DriveShare using email and password authentication.
 - During registration, users must set three security questions (with answers).
- Car Listing and Management (Owner):
 - Car owners can list their vehicles for short-term rental, providing details such as car model, year, mileage, availability calendar, pick up location and rental pricing.
 - Owners have the ability to manage their car listings, including updating availability and price
 - The system must prevent the same car from being rented by more than one renter for overlapping dates/times.
- Search and Booking (Renter):
 - Renters can search for available cars based on location, date, and other preferences (your choice).
 - Renters can “watch” desired cars and receive notification when the car becomes available and/or rental prices drop to meet renter-defined specifications
 - Implement a booking system allowing renters to reserve a car for a specific period.

- Messaging and Communication:
 - DriveShare includes a messaging system to facilitate communication between car owners and renters.
 - Send notifications for booking requests, confirmations, and important updates via email or in-app messages.
- Payment:
 - Allow renters to make payment (A real payment is not required. A payment button with amount is enough, clicking on the button will change the balance and send notification to both owner and renter)
- Rental History (**optional for working on the project alone**)
 - Both car owners and renters can view their rental history.
- Reviews: (**optional for working on the project alone**)
 - Implement a review and rating system for both owner and renter to provide feedback on the rental experience.

Implementation

- User Authentication and Encryption:
 - Implement the Singleton pattern to manage the user's session securely.
- Observer Pattern for Booking Notifications:
 - Apply the Observer pattern to notify renters if their watched cars meet their expectations.
- Mediator Pattern for UI Components:
 - Implement the Mediator pattern to manage communication between different UI components, creating a cohesive and user-friendly interface.
- Builder Pattern for Car Listing Creation:
 - Use the Builder pattern to construct car listing objects with optional/variable attributes cleanly
- Proxy Pattern for Payment Integration:
 - Use the Proxy pattern to simulate secure interaction between DriveShare and the payment system.

- Chain of Responsibility for Password Recovery:
 - o Apply the Chain of Responsibility pattern to create a secure process (using three security questions to build the chain) for recovering a forgotten password.

- **Notes:**
 - o Provide detailed design of the patterns used:
 - Include class diagrams and clearly map:
 - Pattern roles/classes (e.g., Subject/Observer, Mediator/Colleagues, Handler chain, Proxy/RealSubject)
 - Your actual application classes that implement those roles
 - o Your code must be:
 - Thoroughly commented
 - Consistent with the design (diagrams should match the implemented model)

- **Submission**
 - o **Submit One .zip file that contains the followings**
 - **Source code**
 - **A report that includes class diagrams and their descriptions, database schema and descriptions, user-interface screen shots and descriptions, references.**